

Final Report

Biomass-to-Jet: Using Gasification and Fischer–Tropsch Chemistry

Anna Cai, Miguel Piza, Jazmin Aca, Qussay Suleiman

Che 396: Senior Design 1

Professor Betul Bilgin

University of Illinois at Chicago

December 12, 2025

Table of Contents

Introduction	3
Problem Statement	3
Market Analysis	5
Judging Criteria.....	6
Process Description	7
Chosen Process	7
Competing Processes.....	15
Appendix	18
Design Basis	18
Block Flow Diagram	21
Process Flow Diagram	22
Rough material Balance	23
Bibliography	25

Introduction

Problem Statement

Aviation is one of the most energy-intensive transportation modes and is inherently difficult to decarbonize due to strict safety regulations, long asset lifetimes, and the absence of cost-effective alternatives to high-density liquid fuels. Conventional jet fuel is derived from fossil crude oil through distillation and hydrotreating, and its combustion contributes significantly to global greenhouse gas emissions, as well as particulate matter and NO_x, which impact both climate and air quality. Worldwide, aviation accounts for 2% of all carbon dioxide (CO₂) emissions and 12% of all CO₂ emissions from transportation. Global and national policy initiatives are establishing targets to decarbonize the aviation sector. ICAO's CORSIA program caps net CO₂ emissions from international aviation at 2020 levels through 2035, and the aviation industry has set a goal of reaching net-zero emissions by 2050. Because electrification and hydrogen are unlikely to be widely deployed in the near term, Sustainable Aviation Fuel (SAF) represents the most practical pathway to meet these objectives. In the U.S., the 2021 Sustainable Aviation Fuel Grand Challenge aims to scale domestic SAF production to 3 billion gallons annually by 2030 and 35 billion gallons annually by 2050, with a lifecycle reduction in GHG emissions of at least 50%, thereby creating long-term demand and policy support for technologies such as FT-SPK.

Sustainable Aviation Fuel (SAF) is an alternative fuel made from non-petroleum feedstocks such as biomass, waste oils, agricultural residues, and municipal solid waste. SAF has emerged as the most viable pathway for reducing aviation emissions because it is a drop-in fuel that can meet the same performance standards as Jet A without requiring modifications to jet engines, fuel systems, or airports. This is critical for near-term decarbonization because the

physical and regulatory constraints of aviation make it impractical to implement rapid changes in propulsion systems, fuel storage, and aircraft design. SAF must therefore meet stringent performance specifications, including energy density, combustion characteristics, volatility, and cold-flow properties, while offering substantial lifecycle reductions in greenhouse gas emissions.

Multiple production pathways for SAF have been developed and approved under ASTM International standards, including Hydro processed Esters and Fatty Acids (HEFA), Alcohol-to-Jet (ATJ), and Fischer-Tropsch Synthetic Paraffinic Kerosene (FT-SPK). Among these, FT-SPK has gained significant attention due to its ability to convert a diverse range of low-value, sustainable feedstocks into high-quality paraffinic hydrocarbons with ultralow sulfur and aromatic content. When produced from biomass or waste-derived syngas, FT-SPK can provide substantial lifecycle emission reductions and additionally support waste diversion efforts, contributing both environmental and societal benefits.

Despite these advantages, SAF production remains limited by high capital intensity, feedstock logistics, and operational complexity, resulting in production volumes that comprise less than 1% of global jet fuel demand. However, the combination of international emissions mandates, national policy incentives, and long-term corporate sustainability commitments is driving rapid development of the SAF sector. In this context, evaluating the technical, economic, and environmental performance of FT-SPK production is essential to determine optimal process configurations, assess scalability, and identify barriers to commercialization. This project focuses on the design and preliminary analysis of a commercial-scale FT-SPK process capable of supporting the emerging demand for low-carbon aviation fuels.

Market Analysis

The Sustainable Aviation Fuel (SAF) market is poised for significant growth, driven by increasing environmental concerns, regulatory pressures, and advancements in fuel production technologies. Despite its promising potential, SAF production faces challenges, including high capital costs for establishing production facilities, difficulties in securing consistent feedstocks, and operational complexities. These factors contribute to the high production cost of SAF, which currently ranges from \$3.50 to \$6.00 per gallon, significantly higher than traditional jet fuel, which typically costs between \$1.50 and \$2.00 per gallon. This price disparity presents a barrier to widespread adoption. However, as production scales up, technology improves, and government incentives such as tax credits, subsidies, and other financial support are implemented, the price of SAF is expected to decrease. For example, the U.S. SAF Grand Challenge aims to produce 3 billion gallons of SAF annually by 2030, which will help lower production costs through economies of scale and infrastructure development.

The global SAF market is projected to grow from USD 2.7 billion in 2025 to USD 28.6 billion by 2032, at a CAGR of 39.95%. North America is expected to remain the largest market, accounting for 46% of the market share by 2024, largely due to strong government-backed initiatives such as the SAF Grand Challenge. Europe is also a key player, driven by the European Green Deal and national goals for carbon-neutral aviation. Meanwhile, the Asia-Pacific region is set to experience rapid growth, fueled by rising investments, increasing demand for cleaner fuels, and technological advancements in SAF production. The combination of these factors, alongside the scaling of SAF production, is expected to drive down costs and make SAF more competitive with traditional jet fuel, supporting its long-term adoption in the aviation industry.

Judging Criteria

The Fischer-Tropsch (FT) pathway was selected over alternative sustainable aviation fuel production pathways, based on its superior feedstock flexibility, blending capability, scalability, long-term feasibility for meeting the growing SAF demand, and its potential for reducing carbon emissions during the life cycle of the product.

One major advantage of FT is its ability to process a broad range of carbon-based feedstocks like woody biomass, agricultural, Municipal Solid Waste (MSW), etc. The design basis of the plant accounts for this advantage by choosing St. James Parish, Louisiana, where forestry residue is abundant at a low cost. In contrast, other competing processes like HEFA are constrained almost exclusively to rich lipid feedstocks which are highly limited. These feedstock sources are heavily competed for by the renewable diesel market, leaving little to no room for the SAF market. Alcohol to Jet (ATJ) feedstocks depend on sugar, starch, or cellulosic ethanol. Food and agricultural waste have a highly competitive market as these feeds are often used in biofuel industries to produce things like biodiesel. Because FT can utilize non-food, non-lipid, waste biomass at very large scales, it provides a more robust and sustainable feedstock supply chain.

Fisher-Tropsch derived synthetic paraffinic kerosene (FT-SPK), also known as jet fuel, is certified under the ASTM D7566 Annex A1 to be blended at up to 50 vol% with conventional Jet A/A-1 jet fuel. This high blending limit provides two critical advantages. It enables significant displacement of fossil fuel derived jet fuel per unit of SAF produced and allows for drop-in use without any modifications to aircraft engines, fuel handling systems, or airport logistics that are designed to operate with fossil fuel derived kerosene. Other pathways offer a lower maximum blend limit ranging from 10% to 30% thus FT offers both high blending capabilities and feedstock scalability.

The FT pathway also enables deep decarbonization because the biomass to liquids routes achieves significant carbon intensity reductions depending on feedstock logistics and hydrogen sourcing. Other pathways simply do not offer this level of decarbonization due to FT's ability to reduce greenhouse gas emissions.

The long-term expansion potential for FT-SPK is very promising as SAF initiatives and demands continue to increase. Existing commercial plants demonstrate that 10,000 barrels/day is sustainable and efficient making the design of this plant strategic. Competing processes are generally smaller in scale, are limited by supply of feedstocks, and are less aligned with long-term SAF volume requirements mandated by U.S and EU policy. The gulf coast also already supports biomass supply chains, gasification and syngas-handling experience, hydrogen production and contains existing refinery and upgrading systems.

In summary, the Fisher-Tropsch pathway was selected because it provides broad and abundant feedstock options, 50% blend certification, which matches or exceeds alternatives, superior scalability for large volume SAF production, and lower carbon intensity. For these reasons, FT to SAF offers the most robust, sustainable, and scalable pathway for meeting long-term aviation decarbonization targets compared to methods like HEFA or ATJ.

Process Description

Chosen Process

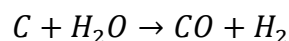
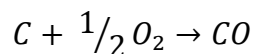
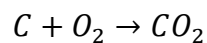
The general Fischer-Tropsch process is as follows. The first step is feedstock selection, which can be any carbon-based material. In this case, wood-based biomass is gasified to create synthetic gas, syngas, which is primarily a mixture of carbon monoxide and hydrogen gas ($\text{CO}_{(g)}$ + $\text{H}_{2(g)}$). The syngas must then undergo cleaning to ensure that a relatively low concentration of

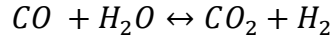
carbon dioxide (CO₂) remains. The lean syngas then undergoes catalytic reactions in an FT reactor that uses cobalt as a catalyst in low temperatures to form long chain hydrocarbons. These hydrocarbons are then sent to upgrading and fractionation processes to produce synthetic paraffinic kerosene (SPK) that meets the ASTM D7566 standard. The final product once blended at up to 50% must meet the new ASTM D1655 standard.

The detailed process will follow the major sections of the process flow diagram highlighted below (Figure 8 - Figure 9) which include biomass gasification and syngas cleanup, Fischer-Tropsch synthesis, hydrotreating, and hydrocracking.

The process begins with the gasification of wood-based biomass to produce syngas which is then cleaned to produce lean syngas that are fed into the FT reactor. For this process, the necessary feed is found in the design basis section to be 7,500 tonnes/day of woody biomass. Unlike fossil-based syngas production, biomass gasification produces syngas containing significant amounts of CO₂, particulates and ash, as well as other contaminants. Extensive gas cleanup and conditioning are required prior to FT synthesis.

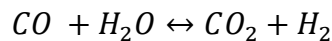
The first step in the biomass gasification process (**Error! Reference source not found.**) is to dry the incoming biomass in a rotary drum dryer (D-100) operating at about 120-180°C to reduce the typical inlet moisture of 30-50 wt.% down to <10 wt.%. The feed is then ready for the biomass gasifier (R-100) where the dried biomass is converted into raw syngas via partial oxidation reactions with air. The primary reactions are shown below (Equation 1).





Equation 1: Primary Biomass Conversion Equations

Assuming a fluidized bed reactor is used then the temperature can be specified at 800-950°C with pressure at 1-3 bar. The products include raw syngas and bottom ash/char. The Cyclone C-100 removes fine ash and char particles from raw syngas to protect the downstream equipment and operates at 700-900°C with a pressure drop of <10 kPa. E-100 then cools the hot syngas to enable downstream wet scrubbing and contaminant removal. The incoming gas will be at around 700-900°C and will leave at ~200°C. Following cooling, the syngas scrubber (V-100) removes remaining contaminants using water and operates at 40-80°C and ~1-3 bar. The knockout drum (V-101) removes entrained liquid droplet and condensed liquids from scrubbed syngas at 40-60°C. The water-gas shift reactor R-101 adjusts the H₂/CO ratio to approximately 2, as mentioned in the design basis and required for cobalt FT synthesis. The main reaction here is shown below (Equation 2). The reactor is expected to operate at high temperatures (350- 450°C) and at 20-30 bar. From the design basis the steam to dry syngas ratio can be taken to be 0.5kg/kg.



Equation 2: Water-Gas Shift Reaction

The final V-102 absorber reduces the concentration of CO₂ to improve FT reactor productivity and operates at 20-30 bar removing 70-90% of CO₂. The resulting lean, conditioned syngas meets the FT feed specifications of having a H₂/CO ratio of 2, particulate concentration < 0.1 mg/Nm³, and negligible concentration of sulfur and nitrogen species.

Key assumptions include that woody residues will have a consistent composition, air-blown gasification is acceptable, heat integration will greatly reduce external utility demands, and overall biomass to syngas efficiency aligns with reported ~70% cold gas efficiency.

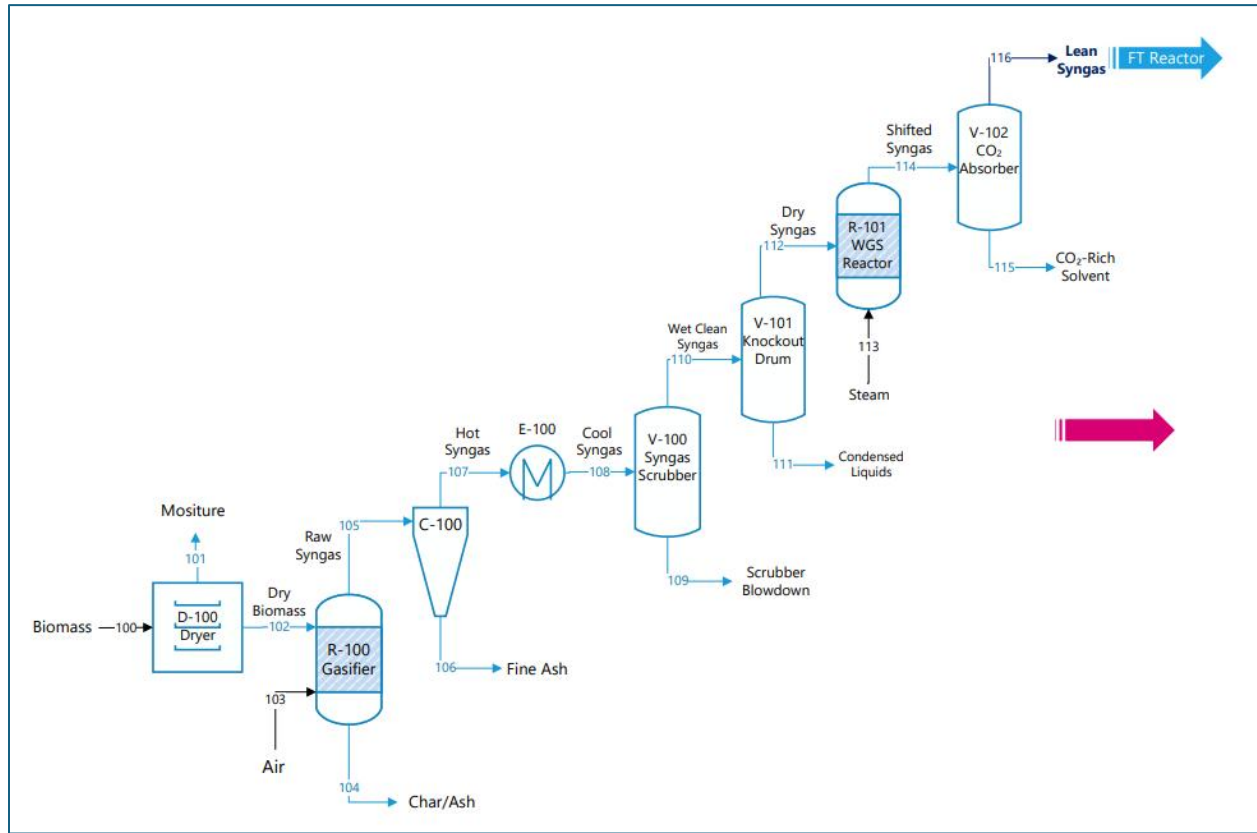
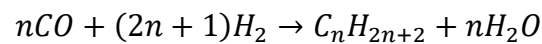
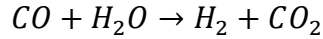


Figure 1: Biomass Gasification and Syngas Cleanup Process Flow Diagram

Following the production of lean syngas, this is fed into the reactor section (Figure 2) of the process. This part of the process converts the conditioned syngas into long chain hydrocarbons producing waxes and middle distillates via the following chemical equations (Equation 13).





Equation 13: Fischer-Tropsch Reactions

The design basis will cover the optimal CO/H₂ ratio in the syngas feed. As previously mentioned, the FT-synthesis will occur in a low temperature fixed bed reactor at ~200-240°C and 20-25 bar with a 20 wt.% cobalt supported catalyst. The FT reactor produces hydrocarbons with a molecular weight distribution which can be described by the Anderson-Schulz-Flory distribution (**Error! Reference source not found.**) .

$$\frac{W_n}{n} = 1 - \alpha^2 \alpha^{n-1}$$

Equation 2 4: Anderson-Schulz-Flory distribution

The catalyst was chosen due to Cobalt's high activity for long-chain paraffins in the desired SAF range of C₈-C₁₆. Alternatives not chosen include iron because it is less selective towards middle distillates, and nickel because of concerns of excessive methane formation. Key assumptions include those of the design basis.

High pressure (HP) and low pressure (LP) flash drums are added to separate heavy waxes (C₂₀⁺) from liquid hydrocarbons and further separate liquid hydrocarbons into naphtha, middle distillates, and light hydrocarbons operating at ~20-25 bar and ~1-5 bar respectively. Wax from the HP drum is sent to hydrocracking and the unreacted syngas from the LP drum are vented. The recycled gas handling units (V-302 & V-303) recycle unconverted syngas back into FT reactor to improve conversion, specifically V-302 separates water and light hydrocarbons and V-303 removes inert gases and flue gas vent.

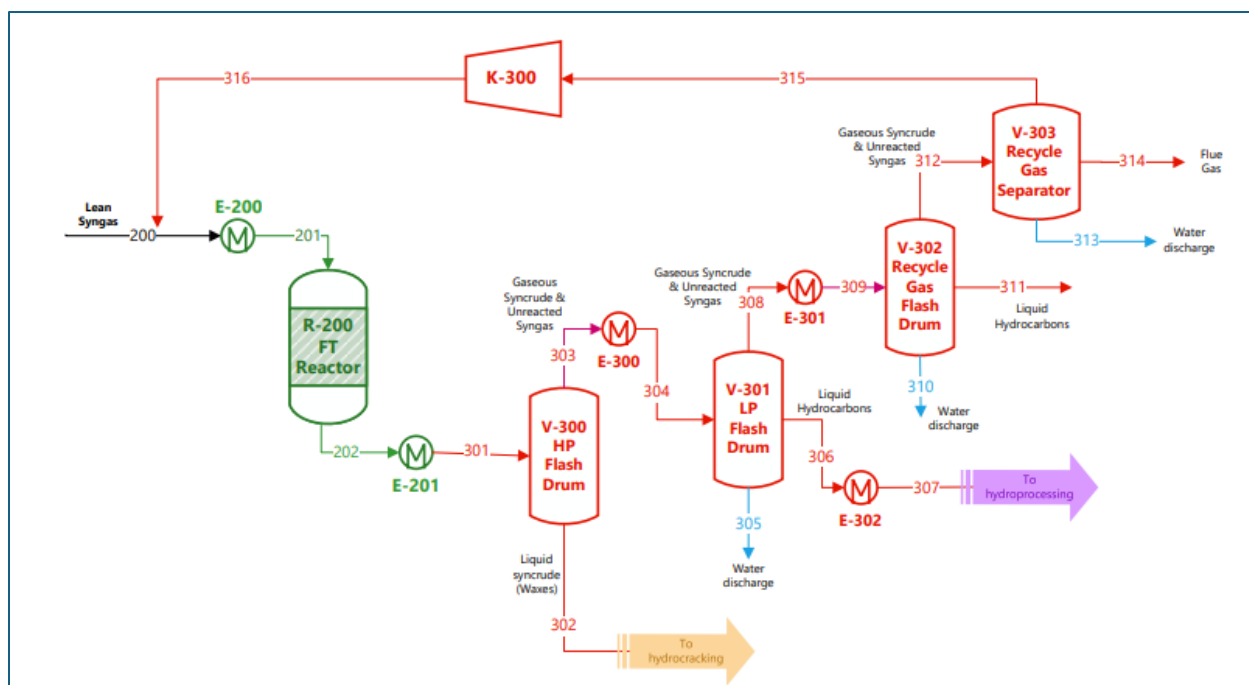


Figure 2: Fisher-Tropsch Synthesis Process Flow Diagram

Following the low temperature cobalt Fisher-Tropsch synthesis the products, which include linear paraffins, with a large fraction of heavy waxes (C_{20}^{+}) and a smaller fraction of middle distillates (C_9-C_{20}) must be sent to upgrading processes like hydrotreating and hydrocracking (Figure 3, Figure 4). While the molecules produced are sulfur and aromatics free, they do not meet aviation fuel specifications due to excessive molecular weight and lack of isomerization. Hydrocracking and hydrotreating are required to crack FT waxes into jet fuel carbon range, isomerize linear paraffins into branched iso-paraffins to meet SAF freezing point limits, and produce co-products, diesel and naphtha.

In the hydrotreating (Figure 3) section R-400 serves as the primary hydrotreating reactor for FT liquids ensuring hydrogenation of olefins and removal of trace oxygenates. The feed includes FT liquids and blue hydrogen. Typical operating conditions include temperatures at 300-380 °C and pressure at 30-50 bar. The flash drum V-400 separates H_2 -rich off gas and

hydrotreated liquid hydrocarbons that can be sent to fractionation. The temperature and pressure are typically in the 200-300°C range. The primary fractionation column C-400 separates naphtha/light ends, middle distillate, and heavy wax/oil bottoms via atmospheric or mild vacuum distillation. The column will typically operate at a top temperature of ~150°C with a bottom temperature at 330-360°C at 1-2 bar. The final fractionation occurs in the finishing column C-401 to separate SAF from heavier diesel-range hydrocarbons. Operating conditions here are similar to refinery kerosene fractionators and main products include SAF and diesel co-product.

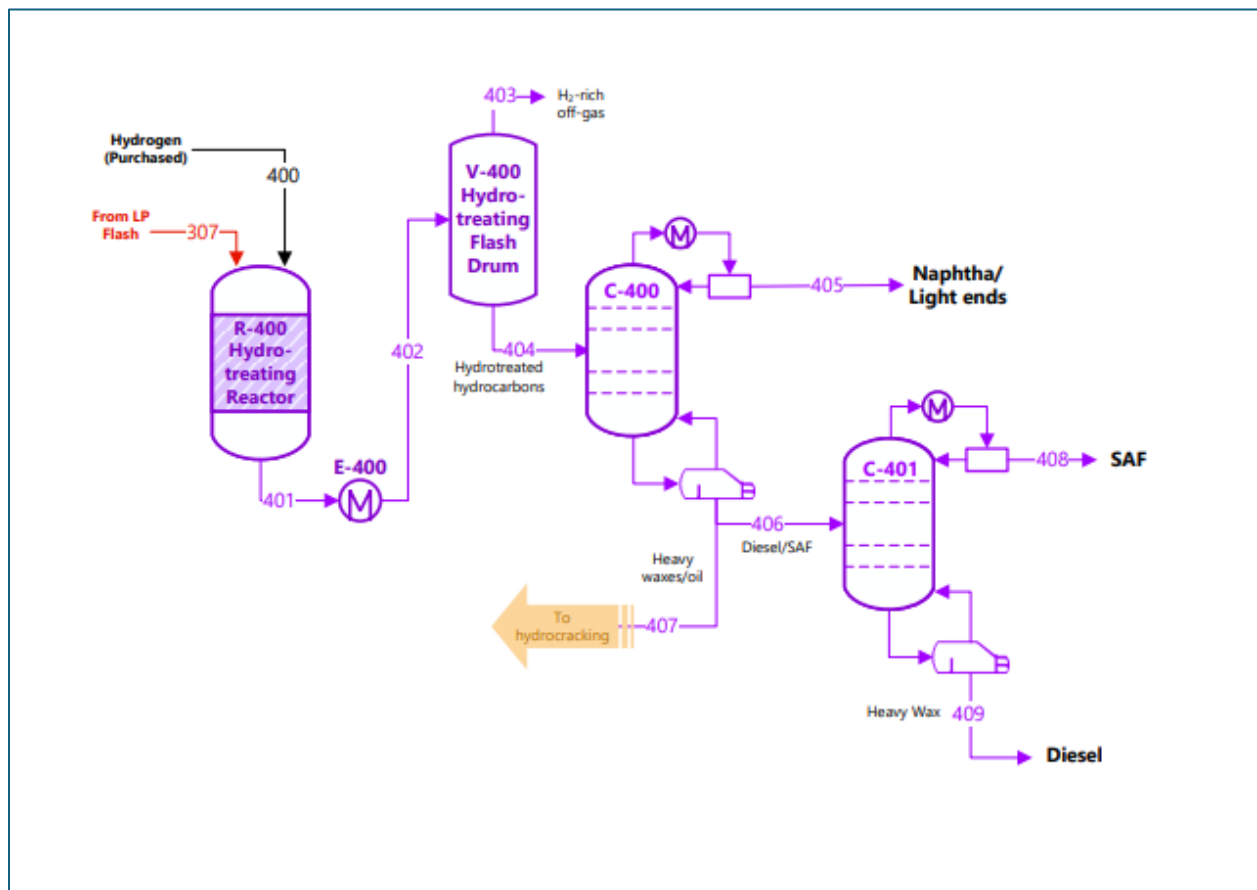


Figure 3: Hydrotreating Process Flow Diagram

The hydrocracking section (Figure 4) serves to crack FT wax into lighter paraffins while simultaneously isomerizing products. Feed streams include heavy waxes from hydrotreating

(Stream 407) and makeup blue hydrogen (stream 500). The hydrocracker typically operates at 350-420°C and 70-150 bar. The conditions are chosen to promote isomerization and high wax conversion. The V-500 flash drum separates flue gas and liquid cracked hydrocarbons at 200-300°C and 50-100 bar. The fractionation columns C-500 & C-501 are a two-step fractionation to recover Naphtha/light ends, SAF, Diesel product, and unconverted heavy wax/oil as a recycle stream. Unconverted waxy material is recycled to maximize SAF yield and overall carbon efficiency.

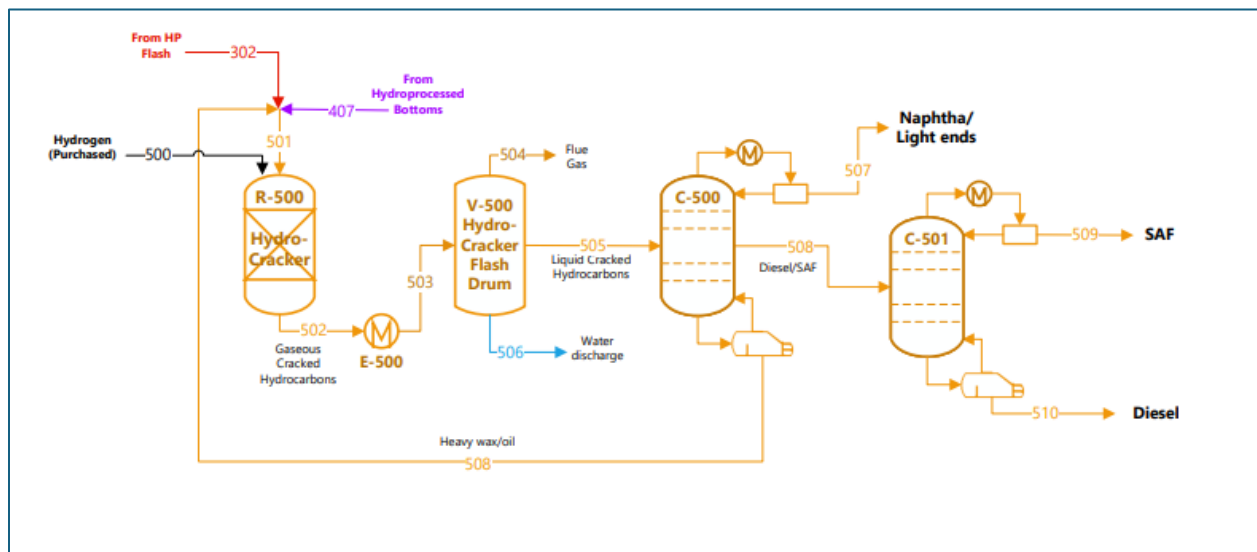


Figure 4: Hydrocracking Process Flow Diagram

Key assumptions in the upgrading section include that FT liquids contain negligible sulfur and nitrogen, eliminating the need for deep hydrotreating. The hydrogen is also assumed to be externally supplied from regional Gulf Coast infrastructure. The hydrocracking severity is tuned to maximize jet fuel range rather than gasoline.

Competing Processes

One of the main competing processes of Fischer-Tropsch process is the Hydro processed Esters and Fatty Acids (HEFA/HVO) process. Like the name suggests, hydro processed esters and fatty acids are the feed stock for this process with hydrotreated vegetable oil as another option. The first step of the process is pretreatment to remove any impurities. Pretreatment is crucial for SAFs especially since unlike traditional crude oil, it contains contaminants like phosphorus and metals. The pretreatment follows 5 steps: feedstock blending, straining, caustic treatment/water washing, impurity adsorption, and vacuum and filtration. Post pre-treatment hydrogenation and hydrodeoxygenation occur. Hydrogenation saturates any double bonds in the fatty acid chains, stabilizing them. Hydrodeoxygenation is where oxygen is removed via the creation of water (H_2O). Both processes are catalyzed, the first is typically by Pd/C, Pt/C, Ru/C or Raney Ni for hydrogenation and Ni, Co, Mo, W and Al_2O_3 as typical catalysts for hydrodeoxygenation.

Following Hydrotreating, the product contains a large concentration of high carbon range materials C15- C18 paraffins, and this material is typically hydrocracked down to a useful range of carbons for SAFs (C8-C16). After the material is cracked there is a large concentration of straight chains that must be converted into branched isomers. The branched isomers have differing physical properties that are desirable for aviation fuel. These isomers control the fuel's ignition time, freezing point, viscosity, density, and flash point all of which must fall within aviation limits. After the isomers are created, they are separated via a high-pressure hot separator, cold high-pressure separator, and then fractionator columns. This leaves the products jet-fuel, diesel or naphtha. After this, any remaining gas product has hydrogen recovered and is then recycled back into the hydrogenation reactor.

The HEFA process is not without its limitations and shortcomings. Primarily it is more economical for these plants to develop renewable diesel over SAFs. Secondly, feedstocks have existing uses such as food production. And finally, it is more rigid in its production of carbon ranges, typically producing only C8-C16. FT synthesis is able to be tailored to a desired range of carbons for various uses. The block flow diagram for a HEFA process is shown below (Figure 5).

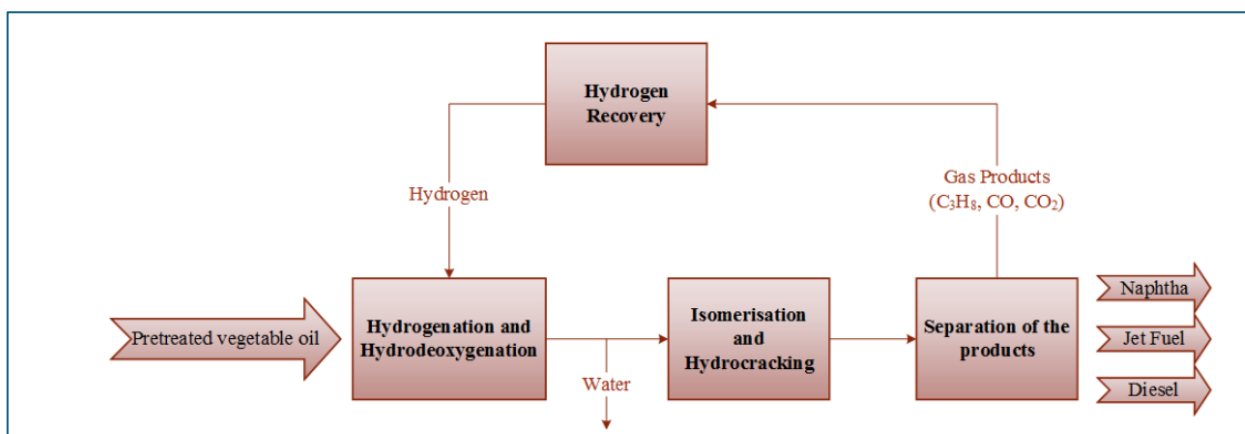


Figure 5: HEFA Block Flow Diagram

Another major competing process for Fischer-Tropsch synthesis is the alcohol to jet fuel (ATJ) route. The first step in the ethanol to jet process is the catalytic fueled dehydration of ethanol to ethylene. Typical catalysts for this process include HZSM-5 which is a zeolite and rare earth metals such as Lanthanum. HZSM-5 is the most common catalyst but is highly acidic, leaving it susceptible to coking. This results in a lower lifetime and large catalyst cost and regeneration rate. As a result, the ATJ process has a large catalyst cost and leaves room for improvements. Following the dehydration, the product is oligomerized. The olefins are then combined into larger carbon chains typically from the C8-16 range. Typical catalysts for this step

include zeolites and Ni-based. The typical choice for catalysts is Ni-ALSBA-15 due to a large conversion of ethylene at 95% and a high selectivity for the C5-C7 range.

Following the oligomerization, the material is often sometimes separated before it is hydrogenated, but this step is not necessarily done in this way. The last step of the process is hydrogenation, and this process is typically done through two separate reactors with two different catalysts. The first reactor typically targets the shorter hydrocarbons in the C8-C11 range while the second reactor achieves a higher conversion of the larger hydrocarbon chains. Typical catalysts for the hydrogenation step include Ni-C and Pt/AL₂O₃. Following hydrogenation, the product is separated via distillation.

Similarly, HEFAs/HVOS alcohol to jet fuel has its limitations. It also is limited in its carbon ranges of C8-C16. It does offer a more flexible feedstock than HEFA but lacks a large commercial scale. It is limited by alcohol availability in each area as alcohol is a high demand product for other industries. Overall FT synthesis offers a large amount of flexibility in product creation and a large amount of feedstock options. For these reasons the decision to go with an FT-based production pathway was made. The block flow diagram for ATJ process is shown below (Figure 6).

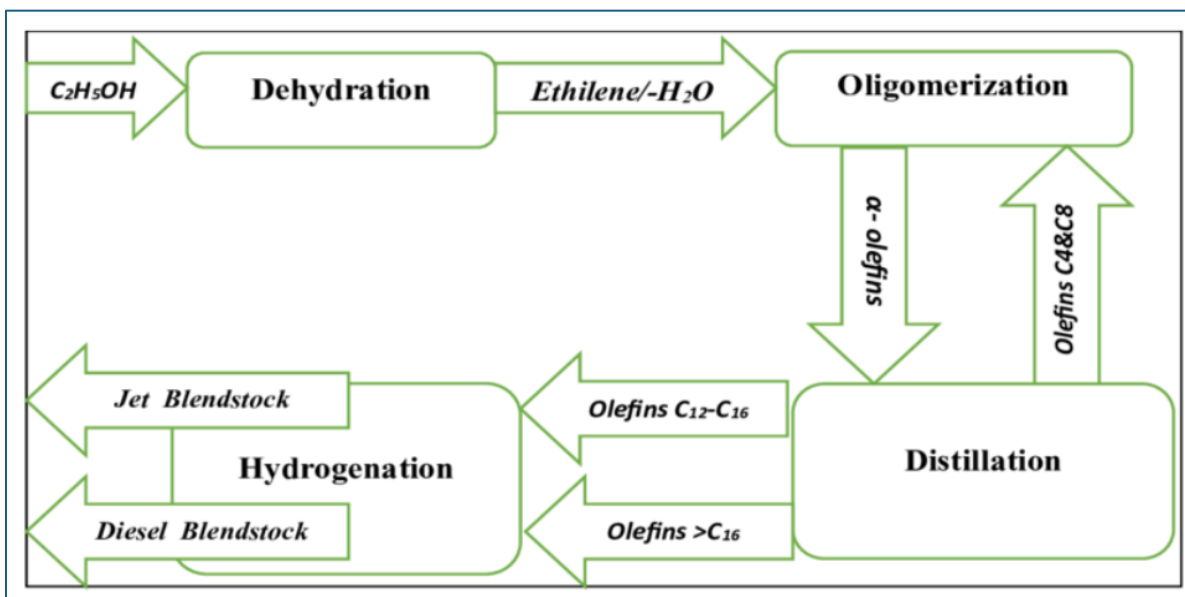


Figure 6: ATJ Block Flow Diagram

Appendix

Design Basis

This design is based on a 10,000 barrels per day (bbl/day) sustainable aviation fuel (SAF) production facility using biomass gasification, syngas conditioning, low-temperature cobalt Fischer–Tropsch (FT) synthesis, and downstream hydrocracking/hydrotreating. The overall configuration and scale are benchmarked against commercial FT-SAF projects under development along the U.S. Gulf Coast—notably the DG Fuels plant in Louisiana, which will employ Johnson Matthey and bp’s FT CANS™ technology at a capacity of 13,000 bbl/day. This provides a realistic industrial reference point for sizing, performance expectations, and catalyst inventory at the scale considered in this study.

St. James Parish, Louisiana is selected as the plant location due to its favorable combination of biomass supply, industrial infrastructure, hydrogen availability, and proximity to

emerging FT-SAF developments. Louisiana ranks among the top U.S. producers of forestry residues, logging waste, and wood-processing by-products, which form the primary biomass feedstock. The Mississippi River petrochemical corridor offers extensive utilities, existing pipelines, and refinery-adjacent infrastructure that reduce capital and operating costs for a biomass-to-liquids (BtL) facility. The region also benefits from strong hydrogen production capacity supported by ammonia and petrochemical operations, as well as growing SAF demand and supportive federal incentives such as the Inflation Reduction Act's SAF tax credits. Deep-water access through the ports of Baton Rouge and New Orleans simplifies logistics for catalyst delivery, equipment transport, and export of finished SAF.

To meet the target of 10,000 bbl/day SAF, the FT synthesis and upgrading section is designed for approximately 1,450 tonnes/day of total FT liquids, of which about 1,250 tonnes/day fall within the SAF carbon range (C_8 – C_{16}), with the remainder as diesel and naphtha co-products. This production scale is consistent with biomass-to-liquids studies reporting overall biomass-to-liquids efficiencies of ~40% (LHV basis) for gasification followed by FT synthesis and upgrading.

At the front end of the process, the plant processes 7,500 tonnes/day of woody biomass, sourced from regional forestry residues and mill by-products. The biomass is dried on-site and fed to the gasifier to produce a raw syngas containing CO, H_2 , CO_2 , steam, and light hydrocarbons.

Biomass gasification naturally produces syngas with a low H_2/CO ratio, which is not suitable for low-temperature cobalt FT catalysts and FT synthesis in general. Because cobalt FT operation requires a target H_2/CO ratio near 2, the design incorporates a water–gas shift (WGS) reactor to partially convert CO and steam to H_2 and CO_2 . Typical design studies report steam-to-

syngas ratios between 0.4 and 0.7 kg steam per kg of dry syngas; here a value of 0.5 kg steam per kg of conditioned syngas is selected, resulting in a steam utility load of approximately 1,000 tonnes/day for syngas conditioning at the required FT reactor feed rate.

The FT synthesis section uses a low-temperature cobalt catalyst, selected for its high activity and strong selectivity toward long-chain linear paraffins, producing heavy wax (C_{20}^{+}) and middle distillate (C_9 – C_{20}) hydrocarbons. Catalyst inventory is scaled from published industrial data indicating that a 100,000 bbl/day FT facility requires roughly 646 tonnes of cobalt. Normalizing to the capacity of this design yields an estimated 65 tonnes of cobalt, corresponding to approximately 320 tonnes of supported catalyst assuming a 20 wt% Co formulation.

Downstream of FT synthesis, the heavy wax and middle-distillate fractions are routed to hydrocracking and hydrotreating units to produce the desired blend of SAF, diesel, and naphtha. The detailed hydrogen consumption associated with these upgrading steps is addressed separately in the mass balance section of the report.

Block Flow Diagram

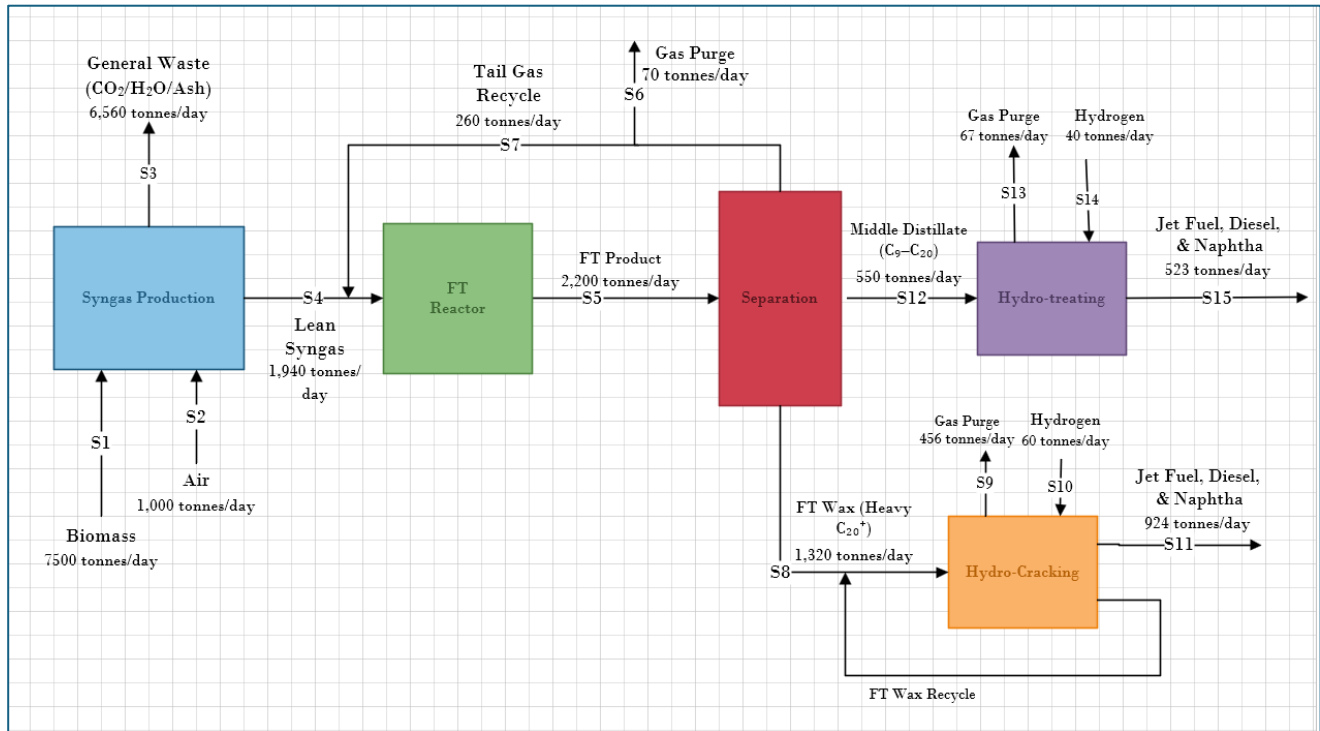


Figure 7: Fischer-Tropsch Synthetic Paraffinic Kerosene Block flow diagram

Process Flow Diagram

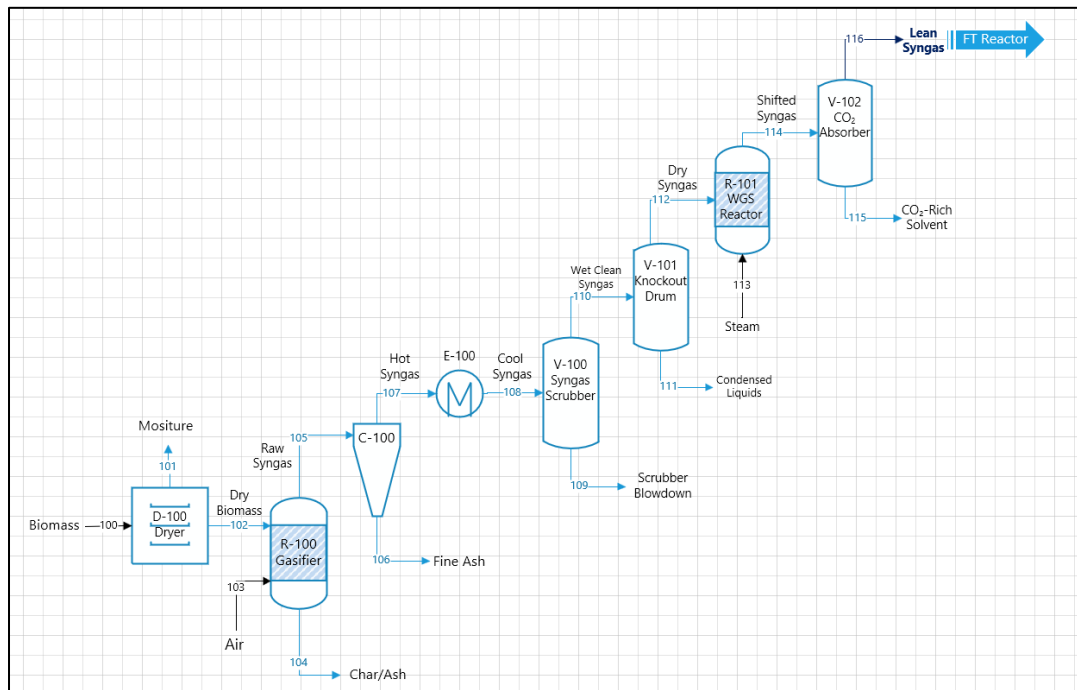


Figure 9: Syngas Production Process Flow Diagram

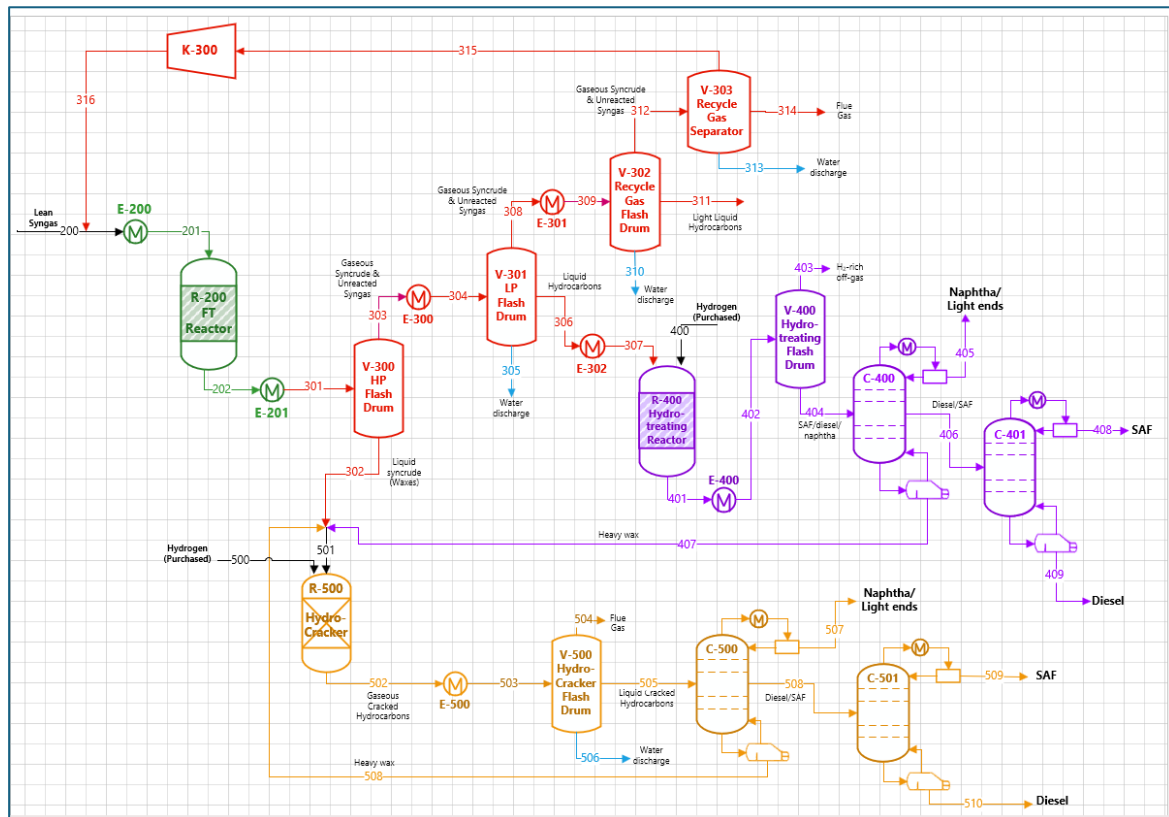


Figure 8: Full FT to SAF Process Flow Diagram

Rough material Balance

The rough material balance will be conducted according to the chosen process' block flow diagram. The major blocks of the process are Syngas Production, FT Reactor, Separation, Hydro-Treating, and Hydrocracking.

Starting from the product target, the liquid section is sized for about 1450 tonnes/day of total FT liquids, with roughly 1250 tonnes/day in the SAF carbon range and the remainder as diesel and naphtha. Working backwards from this target and an overall biomass-to-liquids efficiency near 40 percent on an LHV basis gives a dry biomass requirement of about 7500 tonnes/day. With the biomass-to-syngas conversion set at about 25–30 percent by mass after drying, gasification and cleanup, this corresponds to roughly 1940 tonnes/day of lean syngas to the FT reactor, plus about 1000 tonnes/day of air and a general-waste stream of about 6560 tonnes/day containing CO₂, water, and ash. The FT reactor sees this lean syngas together with a recycled tail-gas stream and converts most of the CO and H₂ into long-chain hydrocarbons. A small FT off-gas purge equal to about 3 percent of the reactor outlet gas is taken to prevent buildup of inerts; the remainder is recycled as tail gas, which closes the FT reactor mass balance.

The liquid FT product flow of about 2200 tonnes/day is split according to the product-distribution assumption: about 60 percent leaves the separator as heavy C₂₀⁺ wax and 40 percent as C₉–C₂₀ middle distillate. The wax stream (about 1320 tonnes/day) is sent to the hydrocracker, where the yield split is set at roughly 70 percent liquids and 30 percent light gas. Together with hydrogen consumption of about 3.5 percent of the wax feed, this gives around 924 tonnes/day of additional liquid fuel, a cracked-gas stream of about 456 tonnes/day, and a hydrocracker purge

sized at about 30 percent of that gas flow. The remaining cracked gas is recycled to the FT section, where it appears as part of the tail-gas recycle. The middle-distillate stream of about 550 tonnes/day is sent to the hydrotreating block, which uses about 1–2 percent of its feed as hydrogen for saturation and cleanup and vents about 5 percent of its reactor effluent as purge gas. The hydrotreating step therefore produces about 523 tonnes/day of upgraded liquids. When the hydrocracker liquids and hydrotreated, middle distillate are combined, the total liquid fuels stream is about 1450 tonnes/day, which matches the design basis for 10,000 barrels per day of finished fuels and provides the 1250 tonnes/day of SAF with co-produced diesel and naphtha. The hydrocracked liquids (S11) and hydrotreated liquids (S15) are later sent through distillation steps to separate the SAF, diesel, and naphtha, but this is only expressed in the process flow diagram. All streams and their descriptions are summarized in the table below (Table 1).

<i>Table 1: Rough Material Balance</i>		
Stream	Description	Mass Flow (tonnes/day)
S1	Biomass (dry basis) to gasifier	7,500
S2	Air to gasifier	1,000
S3	Gasifier waste (CO ₂ /H ₂ O/ash)	6,560
S4	Lean syngas to FT reactor	1,940
S5	FT reactor total hydrocarbon product	2,200
S6	FT section gas purge	70
S7	Tail-gas recycle to FT reactor	260
S8	FT wax (C ₂₀ ⁺) to hydrocracker	1320
S9	Hydrocracker gas purge	456

S10	Hydrogen to hydrocracker	60
S11	Hydrocracker liquid products (SAF/diesel/naphtha)	924
S12	Middle distillate (C ₉ –C ₂₀) to hydrotreating	550
S13	Hydrotreating gas purge	67
S14	Hydrogen to hydrotreating	40
S15	Hydrotreated liquid products (SAF/diesel/naphtha)	523

Bibliography

1. **ASTM International.** *Standard Specification for Aviation Turbine Fuels*; ASTM International: West Conshohocken, PA, 2023.
2. **ASTM International.** *Standard Specification for Aviation Turbine Fuel Containing Synthesized Hydrocarbons*; ASTM D7566-23; ASTM International: West Conshohocken, PA, 2023.
3. **Béalu, Z.; Burger, J.; Friedemann, M.** *Process Simulation and Optimization of Alternative Liquid Fuels Production: A Techno-Economic Assessment of the Production of HEFA Jet Fuel*; Master's Thesis, German Aerospace Center (DLR), 2017.
Available online: https://elib.dlr.de/117806/1/Masterarbeit_Zoe_Bealu.pdf (accessed 2025).
4. **Boymans, E.; Ganjkanlou, Y.; Agirrezabal-Telleria, I.; Viar, N.; Vreugdenhil, B.** Syngas to Sustainable Aviation Fuel: Emerging Catalysts and Routes. *Appl. Catal. A*

2025, 708, 120554.

<https://doi.org/10.1016/j.apcata.2025.120554>.

5. **Consonni, S.; Larson, E.** Biomass-Gasifier/Aeroderivative-GT Combined Cycles: Part A—Performance Analysis. *J. Eng. Gas Turbines Power* **2003**, 125 (3), 745–752.
6. **Curcio, E.** Techno-Economic Analysis of Hydrogen Production: Costs, Policies, and Scalability in the Transition to Net-Zero. *Int. J. Hydrogen Energy* **2025**, 128 (15), 473–487.
<https://doi.org/10.1016/j.ijhydene.2025.04.013>.
7. **Dry, M. E.** The Fischer–Tropsch Process: 1950–2000. *Catal. Today* **2002**, 71 (3–4), 227–241.
8. **Fortunate Business Insights.** *Sustainable Aviation Fuel (SAF) Market Size, Share & COVID-19 Impact Analysis*. Available online:
<https://www.fortunebusinessinsights.com/sustainable-aviation-fuel-saf-market-111560>
(accessed 2025).
9. **Geleynse, S.; Brandt, K.; Garcia-Perez, M.; Wolcott, M.; Zhang, X.** The Alcohol-to-Jet Conversion Pathway for Drop-In Biofuels: Techno-Economic Evaluation. *ChemSusChem* **2018**, 11 (21), 3728–3741. <https://doi.org/10.1002/cssc.201801690>.
10. **Holcombe, T.** *The New CatFT® Process*. Green Impact Fuels, LLC, 2014.
SlideShare. <https://www.slideshare.net/slideshow/catftr-fischer-tropsch-process-presentation/49036312> (accessed 2025).
11. **Idaho National Laboratory.** *Production of Fischer–Tropsch Synfuels at Nuclear Plants*; INL/RPT-22-69047, 2022. Available online:
https://inldigitallibrary.inl.gov/sites/sti/sti/Sort_63673.pdf.

12. **International Civil Aviation Organization (ICAO).** *SAF Production Facilities*.
<https://www.icao.int/SAF/SAF-production-facilities> (accessed 2025).
13. **Johnson Matthey.**
DG Fuels Selects Johnson Matthey and bp's FT CANS™ Technology for Its First Sustainable Aviation Fuel Plant. Available online: <https://matthey.com/products-and-markets/energy/sustainable-aviation-fuel-saf-technology/ft-cans-technology> (accessed 2025).
14. **Light Water Reactor Sustainability Program.** *Production of Fischer–Tropsch Synfuels at Nuclear Plants*; Idaho National Laboratory, 2022.
15. **Meurer, A.; Kern, J.** Fischer–Tropsch Synthesis as the Key for Decentralized Sustainable Kerosene Production. *Energies* **2021**, *14* (7), 1836.
<https://doi.org/10.3390/en14071836>.
16. **Rase, H. F.** *Handbook of Commercial Catalysts: Heterogeneous Catalysts*; CRC Press: Boca Raton, FL, 2000.
17. **Speight, J. G.** *The Chemistry and Technology of Petroleum*, 5th ed.; CRC Press: Boca Raton, FL, 2014.
18. **Trading Economics.** Cobalt Price Data & Historical Charts.
<https://tradingeconomics.com/commodity/cobalt> (accessed 2025).
19. **Trading Economics.** Naphtha Historical Price Data.
<https://tradingeconomics.com/commodity/naphtha> (accessed 2025).
20. **U.S. Department of Energy.** *Sustainable Aviation Fuel Grand Challenge*.
<https://www.energy.gov/eere/bioenergy/sustainable-aviation-fuel-grand-challenge> (accessed 2025).

21. **U.S. Energy Information Administration (EIA).** Gasoline and Diesel Fuel Update.
<https://www.eia.gov/petroleum/gasdiesel/> (accessed 2025).
22. **U.S. Energy Information Administration (EIA).** U.S. Gulf Coast Kerosene-Type Jet Fuel Spot Price. https://www.eia.gov/dnav/pet/hist/eer_epjk_pf4_rgc_dpgD.htm (accessed 2025).
23. **van der Laan, G. P.; Beenackers, A. A. C. M.** *Kinetics and Selectivity of the Fischer–Tropsch Synthesis*. Appl. Catal. A 1999, 193, 39–53
24. **U.S. Energy Information Administration (EIA).**
Louisiana: Wood and Biomass Waste Price and Expenditure Estimates (Table F46).
State Energy Data System (SEDS), 2024. <https://www.eia.gov/state/seds/> (accessed 2025)